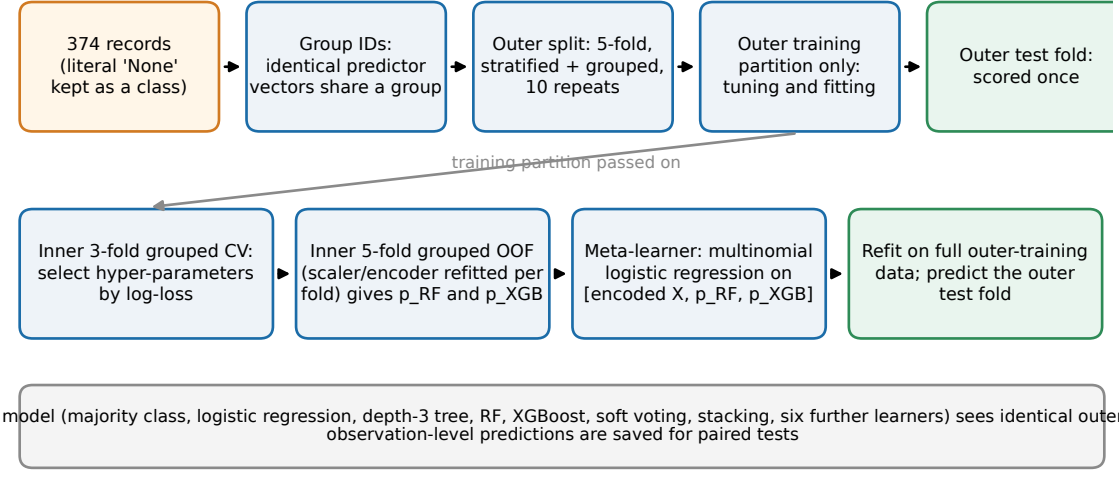
374 records (literal 'None' kept as a class)
Group IDs: identical predictor vectors share a group
Outer split: 5-fold, stratified + grouped, 10 repeats
Outer training partition only: tuning and fitting
Outer test fold: scored once
training partition passed on
Inner 3-fold grouped CV: select hyper-parameters by log-loss
Inner 5-fold grouped OOF (scaler/encoder refitted per fold) gives p_RF and p_XGB
Meta-learner: multinomial logistic regression on [encoded X, p_RF, p_XGB]
Refit on full outer-training data; predict the outer test fold
Every model (majority class, logistic regression, depth-3 tree, RF, XGBoost, soft voting, stacking, six further learners) sees identical outer splits; observation-level predictions are saved for paired tests